Neel Shah

AI Agent Engineering · LLM Tooling · Evaluation Gates

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SUMMARY

I direct AI agents to build and test software, and I build the gates that keep their output honest. CourtVision, a forecasting engine for four sports (NBA, MLB, soccer, tennis), was built mostly by architecting and directing an **agentic workflow**: a **planner model orchestrating cheaper executor models** under hard ship gates. **Claude agents** propose signals and a **statistical gate** rejects them (0 of 60 candidate signal classes shipped; 513 REJECT or DEFER verdicts). An **MCP server** answers queries from Claude with cited artifacts, and an **honesty CI** fails the build when a retracted figure appears outside retraction framing.

SELECTED PROJECT: COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.
Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md

Agentic engineering

- **Agentic build:** built the system mostly by architecting and directing an **agentic workflow**, a **planner model orchestrating cheaper executor models** under hard ship gates.
- **Orchestration playbook:** authored the rules for a **multi-agent** coding loop: **model-tier routing**, parallel **branch-isolated** execution, protected-file guardrails and crash-safe state.
- **Research loop:** an autonomous discovery daemon mines residuals into hypotheses and validates each behind a **statistical gate** (**walk-forward**, null-shuffle permutation control, ablation, Benjamini-Hochberg), with checkpoint and resume; AI agents propose signals and the gate rejects them.
- **MCP and honesty CI:** an **MCP server** (interface documented publicly) answers queries from AI agents with cited artifacts; the build fails when a retracted figure appears outside retraction framing.

Evaluation and guardrails

- **Harness:** expanding-window **walk-forward** with a per-fold date assertion, truncation-invariance leak tests, acceptance only when a change beats baseline on two independent corpora, and a published minimum detectable effect for every market-relative row.
- **Published results:** pregame probabilities match or trail the devigged close and the in-game model is behind the live market, with n and 95% intervals; six withdrawn figures published with reasons at /evidence/retractions/; 16 findings on one template with limitations and verdict history.
- **Pre-registration:** six go-live gates **pre-registered** on 2026-09-17 with thresholds and registration dates in code; execution is paper only and no real order has been placed.
- **Evidence publishing:** a snapshot exporter publishes evidence as a **receipt-backed** bundle (claim, honesty label, source path, sha256), and a **fail-closed** linter blocks the bundle on banned claim language.

Platform

- **Private engine: FastAPI** with ~99 endpoints across 12 routers, 9 long-running daemons under a watchdog, a 430-module codebase, ~7,400-test pytest suite, CI/CD, 5 Docker images; an **MCP server** (interface documented publicly) answers queries from AI agents with cited artifacts.
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing AI coding agents; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Sunsolor · *Tempe, AZ · 2025*
Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist, DataAnnotation.Tech · *Remote · 2024–2025*
Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst, Fortrex Securities · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (99.9% uptime); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

EDUCATION & CERTIFICATIONS

University of Iowa: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.
Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration
Languages: Python (expert), SQL, JavaScript/TypeScript (working)
Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards
ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels